Experiment-12

TCP/UDP Socket Communication

AIM:

To implement end-to-end communication at the Transport Layer using TCP/UDP Sockets and develop a simple chat application.

PROCEDURE:

Echo Client-Server

1. Create a server program.
2. Create a client program.
3. Create a socket at the server.
4. Bind the socket to an IP address and port.
5. Make the server listen for client connections.
6. Accept the client connection.
7. The client sends a message.
8. The server receives and echoes the message.
9. Display the response at the client.

Chat Program

10. Create client and server socket programs
11. Establish a connection.
12. Read & messages from the user.
13. Send the messages through the socket.
14. Receive and display messages at the other end.

Sample Output:

Server

Server started...
Waiting for client...
Client connected.

Received: Hello Server
Echo: Hello Server.

Client

Connected to Server.
Enter message: Hello Server.

Server replied:
Hello Server

15. Continue communication until the user terminates the connection.

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Result:

Thus, end-to-end communication was successfully implemented using socket programming, and echo/chat communication between client and server was verified.